



Revised

Generation Interconnection

Impact Study Report

for

Queue Project AD2-133

EAGLE VALLEY 115KV

18 MW Capacity / 100.33 MW Energy

December 2020

Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between Invenergy Wind Development, LLC, the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Potomac Edison.

Revisions since July 2019 System Impact Study Report

The Dynamic Simulation Analysis section of this report has been revised to align with the Maximum Facility Output (MFO) studied in the Power Flow and Short Circuit Analysis sections. The original Dynamic Simulation Analysis was performed at a reduced output of 5 MW. The results of the Dynamic Simulation Analysis remain unchanged. The update resulted in a slight reduction in the identified reactive power deficiency determined in the Power Factor Assessment.

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM

General

Sandy Ridge Wind, LLC, the Interconnection Customer (IC), has proposed an uprate to an existing wind generating facility located in Centre County, Pennsylvania. This projects requests an increase to the install capability of 100.33 MW with 18 MW of this output being recognized by PJM as capacity. The installed facilities of Q36 & AD2-133 will have a total capability of 150.325 MW with 28 MW of this output being recognized by PJM as capacity. The proposed in-service date for this project is December 31, 2020. **This study does not imply a Mid-Atlantic Interstate Transmission (MAIT) commitment to this in-service date.**

General Information

Interconnection Customer ("IC"): [IC Name] Queue Position: [QUEUE]

Interconnected
Transmission Owner ("TO"): Mid-Atlantic Interstate Transmission, LLC ("MAIT")

Impacted TO(s)
(if applicable): Mid-Atlantic Interstate Transmission, LLC ("MAIT")

PJM Zone: Penelec

FE Operating Company or
Planning Region: Penelec

Customer Connection Request

Requested Backfeed Date: n/a Requested Commercial
Operation Date: 12/31/2020
This study does not imply a FirstEnergy commitment to these dates.

New Facilities		Existing Facilities	
Capacity:	18 MW	Capacity:	10 MW
Energy:	100.325 MW	Energy:	50 MW
MFO ¹ :	150.325 MW	MFO:	50 MW
Fuel:	Wind	Prior Queue Position(s):	Q36

Point of Interconnection

Primary Point of Interconnection: [PRI POI VOLTAGE] kV Substation

¹ Maximum Facility Output

Queue Number	AD2-133
Project Name	EAGLE VALLEY 115KV
Interconnection Customer	Sandy Ridge Wind, LLC
State	None
County	Centre
Transmission Owner	PENELEC
MFO	150.3
MWE	100.33
MWC	18
Fuel	Wind
Basecase Study Year	2021

Point of Interconnection

AD2-133 will interconnect with the MAIT transmission system at the Eagle Valley 115kV substation.

Cost Summary

The AD2-133 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$0
Direct Connection Network Upgrade	\$0
Non Direct Connection Network Upgrades	\$0
Allocation for New System Upgrades	\$607,900
Contribution for Previously Identified Upgrades	\$0
Total Costs	\$607,900

Transmission Owner Scope of Work

Attachment Facilities

No Attachment Facilities are required to support this interconnection request.

Direct Connection Cost Estimate

No Direct Connection Facilities are required to support this interconnection request.

Non-Direct Connection Cost Estimate

No Non-Direct Connection Facilities are required to support this interconnection request.

Incremental Capacity Transfer Rights (ICTRs)

Will be determined at a later study phase

Transmission Owner Analysis

Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2021 summer peak load flow model and the results were verified by FE. Additionally, FE performed an analysis of its underlying transmission <100 kV system. At the Primary POI, the [QUEUE] project contributes to overloads on the FE transmission <100 kV system as shown in Attachment 3a. The estimated cost of system reinforcements necessary to mitigate these overloads are provided in Attachment 3b.

FE Summer Peak Transmission <100 kV Analysis

Contingency Description	Overloaded Element	Rating (MVA)	% Loading After [QUEUE]	Final % Loading	[QUEUE] MW Contrib.	FE Comments/Reinforcements
Bear Rock – Johnstown 230 kV Line & Greenwood – Westfall 46 kV Line	Eldorado – Hydracon Tap 46 kV Line	44	113.00	113.00	7.72	Upgrade terminal equipment at Eldorado substation. Replace relaying at Eldorado and Westfall substations.
Bear Rock – Johnstown 230 kV Line & Eldorado – Summit 46 kV Line	Eldorado – Hydracon Tap 46 kV Line	44	103.54	103.54	5.21	Upgrade terminal equipment at Eldorado substation. Replace relaying at Eldorado and Westfall substations.
Bear Rock – Johnstown 230 kV Line & Eagle Valley – Philipsburg 115 kV Line	Eldorado – Hydracon Tap 46 kV Line	44	102.75	102.75	13.19	Upgrade terminal equipment at Eldorado substation. Replace relaying at Eldorado and Westfall substations.

Eldorado – Westfall 46 kV Line: Upgrade terminals

SS/LN	Estimate No.	Description	Total with Tax	Tax	Total Cost
SS	PN-S-1974	Eldorado-Westfall 46kV line loadability increase for the AD2-133 System Upgrade @ Eldorado SS	330,300	55,900	274,400
SS	PN-S-1975	Eldorado-Westfall 46kV line loadability increase for AD2-133 System Upgrade @ Westfall SS	283,700	48,000	235,700
	EOC	Project Management, Commissioning, and SCADA.	117,800	20,000	97,800
Totals			731,800	123,900	607,900

Estimated Time to Complete: 7 Months

Interconnection Customer Requirements

System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

The IC has requested a non-standard GSU transformer winding configuration. This transformer is in violation of section 14.2.6 of FE's "Requirements for Transmission Connected Facilities" document. This standard requires the GSU transformer to have a grounded wye connection on the high (utility) side and a delta connection on the low (generator) side. However, because AD2-133 is studied as an increase to the output to the existing Q36 project, which was constructed with a GSU transformer winding configuration that does not meet FE's current winding configuration requirements, FE will allow a one-time exception to this requirement and permit the IC to install a GSU transformer with grounded wye windings on both the high and low sides.

Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. If not already existing, the IC is responsible for the following:

1. The purchase and installation of a fully rated [PRI POI VOLTAGE] kV circuit breaker to protect the Q36/[QUEUE] generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the Q36/[QUEUE] generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits.

Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

Power Factor Requirements

The existing 50 MW portion of the Customer Facility associated with queue project Q36 shall retain its existing ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the Point of Interconnection and the increase of 100.325 MW to the Customer Facility associated with [QUEUE] project shall be designed with the ability to maintain a power factor of at least 1.0 (unity) to 0.95 lagging (supplying VARs) measured at the Point of Interconnection.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

PENELEC Requirements

The Interconnection Customer will be required to comply with all FE Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "FirstEnergy Requirements for Transmission Connected Facilities" document located at the following links:

<http://www.pjm.com/planning/design-engineering/to-tech-standards.aspx>

Network Impacts

The Queue Project AD2-133 was evaluated as a 100.33 MW (Capacity 18 MW) injection at the Eagle Valley 115kV substation in the PENELEC area. Project AD2-133 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD2-133 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Light Load Analysis

Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

Affected Systems

LG&E

LG&E Impacts to be determined during later study phases (as applicable).

MISO

MISO Impacts to be determined during later study phases (as applicable).

TVA

TVA Impacts to be determined during later study phases (as applicable).

Duke Energy Progress

Duke Energy Progress Impacts to be determined during later study phases (as applicable).

NYISO

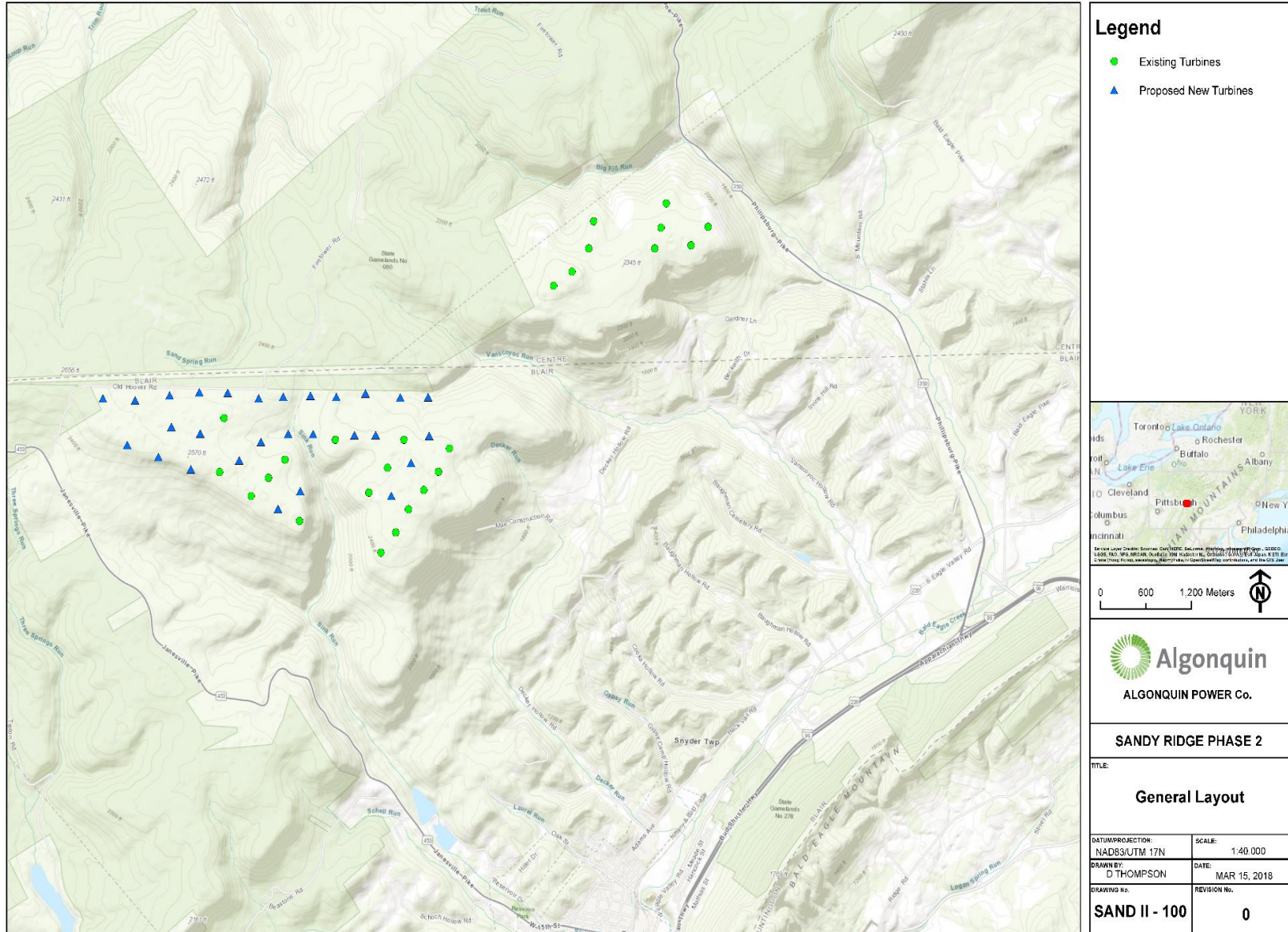
NYISO Impacts to be determined during later study phases (as applicable).

Short Circuit

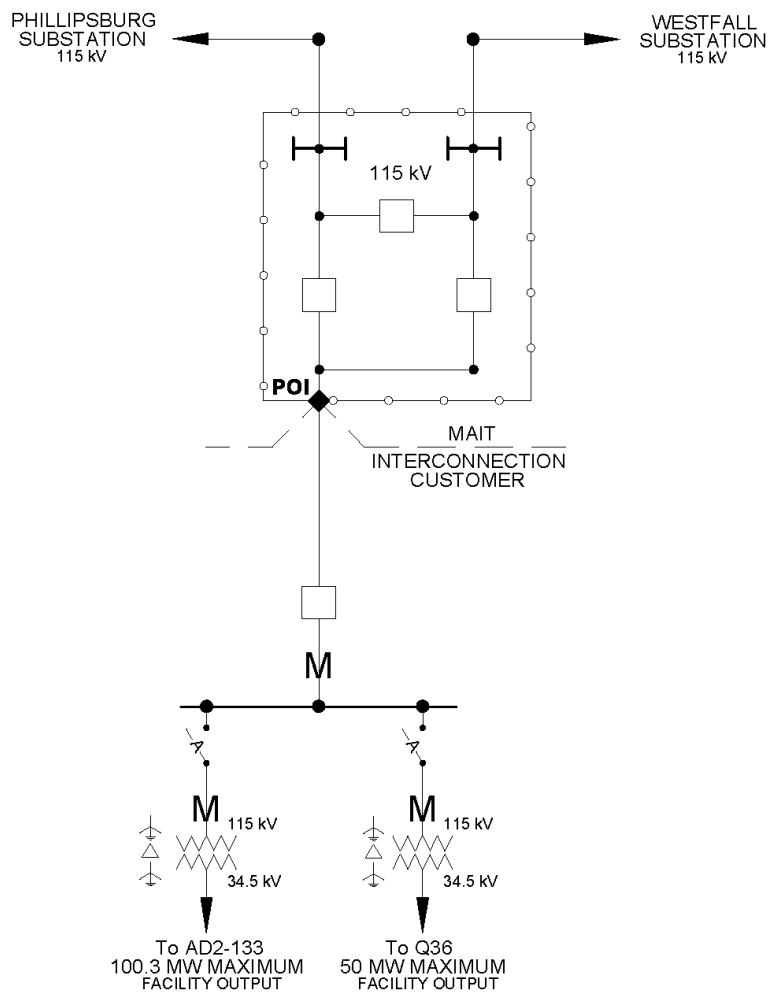
Short Circuit

PJM performed a short circuit analysis and the results were verified by FE. The connection of the AD2-133 project to the system does not result in any newly overdutied circuit breakers on the FE transmission system and does not have a significant fault current contribution to existing overdutied circuit breakers.

Attachment 1. Project Location



Attachment 2. AD2-133 One Line Diagram



◆ = POI LOCATED WHERE INTERCONNECTION CUSTOMER'S BUS CROSSES SUBSTATION FENCE

M = REVENUE METERING FOR INTERCONNECTION CUSTOMER IS OWNED, OPERATED, AND MAINTAINED BY INTERCONNECTION CUSTOMER

FirstEnergy Energy Delivery Technical Services		SANDY RIDGE WIND, LLC SUBSTATION INTERCONNECTION	
BY: J L M	DATE: 03/12/2019	AGREEMENT	DOC ID
APP: -	ISSUE: FINAL		REV: -
		POI-MAIT-PN-AD2-133.dwg	

Attachment 3. Dynamic Simulation Analysis

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Executive Summary

Generator Interconnection Request AD2-133 is for 100.33 MW Maximum Facility Output (MFO) wind generation plant. AD2-133 consists of 6 x 2.625 MW Gamesa G114 and 19 X 4.5 MW Siemens Gamesa SG145 wind turbines with a Point of Interconnection (POI) on the existing Eagle Valley 115 kV substation, in the Pennsylvania Electric (PENELEC) transmission system, Centre County, Pennsylvania.

This report describes a dynamic simulation analysis of AD2-133 as part of the overall system impact study. The load flow scenario for the analysis was based on the RTEP 2021 peak load case, modified to include applicable queue projects. AD2-133 has been dispatched online at maximum power output, with unity power factor and approximately 1.0 pu voltage at the generator terminals.

AD2-133 was tested for compliance with NERC, PJM, Transmission Owner and other applicable criteria. 45 contingencies were studied, each with a 20 second simulation time period (with 1.0 second initial run prior to any events). Studied faults included:

- a) Steady state operation (Category P0);
- b) Three phase faults with normal clearing time on the intact network (Category P1);
- c) Single phase to ground faults with delayed clearing due to a stuck breaker (Category P4);
- d) Single phase to ground faults with delayed clearing as a result of protection failure (Category P5).

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For all 45 fault contingencies tested on the 2021 peak load case:

- a) AD2-133 was able to ride through the faults (except for faults where protective action trips a generator(s)).
- b) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required for the studied project.

Based on the Impact Study data provided, AD2-133 does not meet 0.95 lagging reactive power requirement at the POI. An estimated 5.66 MVAR is needed to meet the lagging power factor requirement.

1. Introduction

Generator Interconnection Request AD2-133 is for 100.33 MW Maximum Facility Output (MFO) wind generation plant. AD2-133 consists of 6 x 2.625 MW Gamesa G114 and 19 X 4.5 MW Siemens Gamesa SG145 wind turbines with a Point of Interconnection (POI) on the existing Eagle Valley 115 kV substation, in the Pennsylvania Electric (PENELEC) transmission system, Centre County, Pennsylvania.

This analysis is effectively a screening study to determine whether the addition of AD2-133 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AD2-133 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

Generator Interconnection Request AD2-133 is for 100.33 MW Maximum Facility Output (MFO) wind generation plant. AD2-133 consists of 6 x 2.625 MW Gamesa G114 and 19 X 4.5 MW Siemens Gamesa SG145 wind turbines with a Point of Interconnection (POI) on the existing Eagle Valley 115 kV substation, in the Pennsylvania Electric (PENELEC) transmission system, Centre County, Pennsylvania.

The project AD2-133 is an uprate to an existing wind generating facility of Q36. The installed facilities of Q36 and AD2-133 will have a total capability of 150.33 MW. The connection diagram of the AD2-133 generating facility is shown in Figure 1.

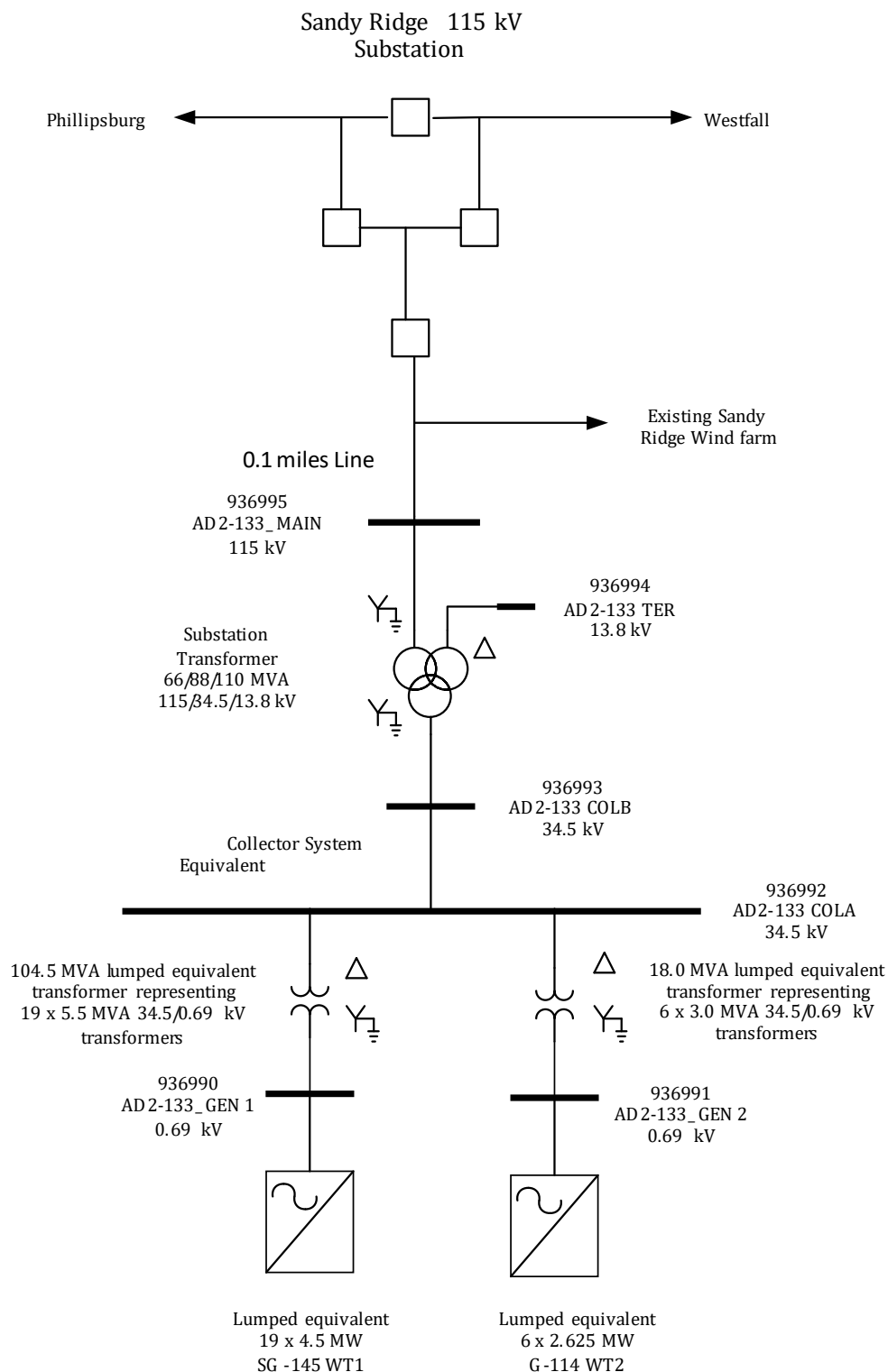


Figure 1: AD2-133 Plant Model

Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AD2-133 loadflow model.

Additional project details are provided in Attachments 1 through 4:

- Attachment 1 contains the Impact Study Data which details the proposed AD2-133 project.
- Attachment 2 shows the one line diagram of the PENELEC network in the vicinity of AD2-133.
- Attachment 3 provides a diagram of the PSS/E model in the vicinity of AD2-133.
- Attachment 4 gives the AD2-133 PSS/E loadflow and dynamic models of the AD2-133.

Table 1: AD2-133 Plant Model

	Impact Study Data	Model
Wind Turbines Lumped Equivalent (WT1)	19 x 4.5 MW SG-145 5.0 MVA Wind Turbines MVA base = 5.0 MVA Vt = 0.69 kV Stator resistance = 0 Saturated sub-transient reactance = 0.131 p.u	1 Lumped equivalent representing 19 x 4.5 MW SG-145 5.0 MVA Wind Turbines Pgen 85.5 MW Pmax 85.5 MW Pmin 0 MW Qmax 41.401 Mvar Qmin -41.401 Mvar Mbase 95.00 MVA R 0 p.u X 0.131 p.u
Wind Turbines Lumped Equivalent (WT2)	6 x 2.625 MW G-114 3.0 MVA Wind Turbines MVA base = 3.0 MVA Vt = 0.69 kV Stator resistance = 0 Saturated sub-transient reactance = 0.131 p.u	1 Lumped equivalent representing 6 x 2.625 MW G-114 3.0 MVA Wind Turbines Pgen 15.75 MW Pmax 15.75 MW Pmin 0 MW Qmax 5.172 Mvar Qmin -5.172 Mvar Mbase 15.75 MVA R 0 p.u X 0.131 p.u
Inverter Based Step-Up Transformer (WT1)	19 x 34.5/0.69 kV wind turbine step-up transformers Rating = 5.5 MVA Transformer MVA base = 5.5 MVA Impedance = 0.009500 + j 0.1140 p.u. Number of taps = N/A Tap step size = N/A	1 x Lumped equivalent representing 19 x 34.5/0.69 kV wind turbine step-up transformers Rating = 104.5 MVA Transformer MVA base = 104.5 MVA Impedance = 0.009500 + j 0.1140 p.u. Number of taps = N/A Tap step size = N/A

	Impact Study Data	Model
Inverter Based Step-Up Transformer (WT2)	6 x 34.5/0.69 kV wind turbine step-up transformers Rating = 3 MVA Transformer MVA base = 3 MVA Impedance = $0.005800 + j 0.0696$ p.u. Number of taps = N/A Tap step size = N/A	1 x Lumped equivalent representing 6 x 34.5/0.69 kV wind turbine step-up transformers Rating = 18.0 MVA Transformer MVA base = 18.0 MVA Impedance = $0.005800 + j 0.0696$ p.u. Number of taps = N/A Tap step size = N/A
Inverter Based Main Transformer (MPT-1)	1 x 115/34.5/13.8 kV Rating = 66/88/110 MVA Transformer MVA base = 66 MVA Impedance: High to Low = $0.0028 + j 0.0899$ p.u High to Tert = $0.0062 + j 0.1169$ p.u Low to Tert = $0.0034 + j 0.0271$ p.u Number of taps = N/A Tap step size = N/A	1 x 115/34.5/13.8 kV Rating = 66/88/110 MVA Transformer MVA base = 66 MVA Impedance: High to Low = $0.0028 + j 0.0899$ p.u High to Tert = $0.0062 + j 0.1169$ p.u Low to Tert = $0.0034 + j 0.0271$ p.u Number of taps = N/A Tap step size = N/A
Auxiliary Load	1.00 MW + 0.1 MVar at low side of the GSU	1.00 MW + 0.1 MVar at low side of the GSU (Switched Offline)
Station Load	0.00 MW + 0.00 MVar at low side of the GSU	N/A
Collector System Equivalent (Collector 1)	34.5 kV line Impedance = $0.01447 + j 0.01719$ p.u Charging susceptance = $j 0.04323$ p.u (Impedance on 100 MVA Base)	34.5 kV line Impedance = $0.01447 + j 0.01719$ p.u Charging susceptance = $j 0.04323$ p.u (Impedance on 100 MVA Base)
Attachment Line	0.1 mile 115 kV line Impedance = $0.00 + j 0.00$ p.u Charging susceptance = $j 0.0$ p.u (Impedance on 100 MVA Base)	0.1 mile 115 kV line Impedance = $0.00 + j 0.00$ p.u Charging susceptance = $j 0.0$ p.u (Impedance on 100 MVA Base)

3. Reactive Power Assessment

AD2-133 was assessed for compliance with reactive power capability requirements using the supplied capability curves. Please note this is a new facility.

- Generation shall have the ability to maintain a power factor of at least 0.95 leading to 0.95 lagging at the POI or the result of the System Impact Study indicated that, for the safety and reliability of the Transmission System, no power factor requirement is required^{2, 3}.

Generator	MFO	Required pF Range		Maximum (Lagging)	Minimum (Leading)
		Lagging	Leading		
AD2-133	100.33	0.95	0.95		
Total MVAR Required				32.98	-32.98
MVAR from Generators				Qmax	Qmin
				46.57	-46.57
Customer Planned Compensation				0	0
Qloss				-19.26	-19.26
Total Available MVAR at POI				27.3	-65.8
Deficiency in MVAR				5.66	Meet

Based on the Impact Study Data provided AD2-133 **does not** meet the reactive power requirement at POI. The estimated required additional capacitive reactive power is 5.66 MVAR to fulfill the power factor requirement.

² As specified in the document "Reactive Power Requirements.doc", Date: 6/15/2018.

³ As specified in Attachment O of the document "PJM Open Access Transmission Tariff" Effective Date: 4/23/2018.

4. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process⁴.

The selected load flow scenario is the RTEP 2021 peak load case with the following modifications:

- a) Addition of all applicable queue projects prior to AD2-133.
- b) Addition of AD2-133 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AD2-133.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AD2-133 initial conditions are listed in **Table 2**, indicating maximum power output, with approximately unity power factor and 1.0 pu voltage at the generator terminals. The POI voltage is at 1.0174 pu (per the PJM scheduled voltage at 115 kV).

Table 2: AD2-133/Q36 machine initial conditions

Bus	Name	Unit	PGEN	QGEN	ETERM	POI Voltage
936991	AD2-133_GEN1	1	85.50 MW	0.16 MVar	1.00 pu	1.0174 pu
936992	AD2-133_GEN2	1	15.75 MW	2.09 MVar	1.00 pu	
200905	26SANDYRIDGE	W1	50.00 MW	1.89 MVar	1.02 pu	

Generation within the vicinity (within five buses away) of AD2-133 has been dispatched online at maximum output (P_{MAX}). The dispatch within the PENELEC area is given in Attachment 5.

⁴ Manual 14B: PJM Region Transmission Planning Process, Rev 37, April 28 2017, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

5. Fault Cases

Table 5 to **Table 8** list the contingencies that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval (with 1.0 second initial run prior to any events).

The studied contingencies include:

- a) Steady state operation (Category P0);
- b) Three phase faults with normal clearing time on the intact network (Category P1);
- c) Single phase to ground faults with delayed clearing due to a stuck breaker (Category P4);
- d) Single phase to ground faults with delayed clearing as a result of protection failure (Category P5).

There is no common tower contingencies (P7) in the vicinity of AD2-133.

High Speed Reclosing (HSR) is used for the following PENELEC 115 kV lines in the vicinity of AD2-133⁵:

- 115 kV circuit EVW (Westfall to Eagle Valley) recloses with 1.0 second delay
- 115 kV circuit TFW (Westfall to Thirty First St.) recloses with 0.4 second delay
- 115 kV circuit SF (Philipsburg to Shawville) recloses with 0 second delay
- 115 kV circuit EVP (Philipsburg to Eagle Valley) recloses with 0 second delay
- 115 kV circuit SD (Shawville to Dubois) recloses with 0 second delay
- 115 kV circuit SM (Shawville to Garman) recloses with 0 second delay
- 230 kV circuit MAN-SHW (Shawville to Moshannon) recloses with 0 second delay
- 230 kV circuit ELK-SHW (Shawville to Elko) recloses with 0 second delay
- 230 kV circuit SHW-SNT (Shawville to Shingletown) recloses with 0 second delay

Buses at which the faults listed above will be applied are:

- AD2-133 POI 115 kV (i.e., Eagle Valley 115 kV)
- Philipsburg 115 kV
- Westfall 115 kV
- Shawville 230/115 kV

Table 3 gives the details of worst fault clearing time⁶ for 115 kV and 230 kV breakers at PENELEC.

Table 3: 115 kV and 230 kV breaker details

TO	Circuit Breaker	Three Phase Fault Normal Clearing Time (cycles)	SLG Delayed Clearing Time due to Stuck Breaker (cycles)	SLG Delayed Clearing Time due to Primary Relay Failure (cycles)
PENELEC	230 kV	5	15*	60

⁵ PJM_HighSpeedReclosing_List.xlsx

⁶ Rev. 20 of "2017 Revised Clearing time for each PJM company_Rev20.xls"

PENELEC	115 kV	7	20	80
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* Shawville is a generating substation

A complete list of the contingencies that will be studied is given in **Table 5** to **Table 8**.

The positive sequence fault impedances for single line to ground faults were derived through the procedure described in *SLG fault equivalent impedance estimation for stability study*, dated 02/10/17.

6. Evaluation Criteria

This study is focused on AD2-133, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) AD2-133 is able to ride through the faults (except for faults where protective action trips a generator(s)),
- b) The system with AD2-133 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 3%.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. Summary of Results

Plots from the dynamic simulations are provided in Attachment 6, with results summarized in Table 5 through **Table 8**.

For all 45 fault contingencies tested on the 2021 peak load case:

- a) AD2-133 was able to ride through the faults (except for faults where protective action trips a generator(s)).
- b) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

Network non-convergence was also observed as summarized in **Table 4**.

Table 4: Summary of Network Non-Convergence

Contingency	Time [s]	Unit	Bus
P1.01*	1.0	AD2-133 GEN2	936991
	1.0167~1.1292	AD2-133 GEN1	936990
P1.02*	1.0	AD2-133 GEN2	936991
	1.0167~2.1292	AD2-133 GEN1	936990
P1.03*	1.0	AD2-133 GEN2	936991
	1.0208, 1.025	AD2-133 GEN1	936990
P1.04*	1.0	AD2-133 GEN2	936991
	1.0208, 1.025	AD2-133 GEN1	936990
P1.05*	1.0	AD2-133 GEN2	936991
	1.0208, 1.025	AD2-133 GEN1	936990
P1.06*	1.0	AD2-133 GEN2	936991
	1.0208, 1.025	AD2-133 GEN1	936990
P1.07*	1.0	AD2-133 GEN2	936991
	1.0042, 1.0125	AD2-133 GEN1	936990
P1.08*	1.0	AD2-133 GEN2	936991
	1.0042, 1.0125, 1.4	AD2-133 GEN1	936990
P1.09*	1.0	AD2-133 GEN2	936991
	1.0042, 1.0125	AD2-133 GEN1	936990
P1.10*	1.0	AD2-133 GEN2	936991
	1.0042, 1.0125	AD2-133 GEN1	936990
P1.11*	1.0042, 1.0125	AD2-133 GEN1	936990
P1.12*	1.0042, 1.0125	AD2-133 GEN1	936990
P1.13*	1.0042, 1.0125	AD2-133 GEN1	936990
P1.14*	1.0042, 1.0125	AD2-133 GEN1	936990
P1.15*	1.0042, 1.0125	AD2-133 GEN1	936990

Contingency	Time [s]	Unit	Bus
P4_1B1.01*	1.1208, 1.1292	AD2-133 GEN1	936990
P4_1B1.02*	1.1208, 1.1292	AD2-133 GEN1	936990
P4_1B1.04*	1.1292	AD2-133 GEN1	936990
P5.01*	1.1208, 1.1292	AD2-133 GEN1	936990
P5.02*	1.1208	AD2-133 GEN1	936990
P5.03*	1.125, 1.1292	AD2-133 GEN1	936990

* Non-convergence during fault that should not be a problem if it recovers back after the fault is removed.

Table 5: Steady State Operation – Category P0

Fault ID	Duration	AD2-133 No Mitigation
P0_01	Steady state 20 sec	Stable

Table 6: Three-phase Faults With Normal Clearing – Category P1

Fault ID	Fault description	Clearing Time & Reclose (Cycles)	AD2-133 No Mitigation
P1.01	3ph fault @ Eagle Valley 115 kV on Philipsburg circuitEVP, normal clear loss of Eagle Valley – Philipsburg 115 kV circuitEVP	7/0	Stable
P1.02	3ph fault @ Eagle Valley 115 kV on Westfall circuitEVW, normal clear loss of Eagle Valley – Tyrone North – Westfall 115 kV circuitEVW	7/60	Stable
P1.03	3ph fault @ Philipsburg 115 kV on Eagle Valley circuitEVP, normal clear loss of Eagle Valley – Philipsburg 115 kV circuitEVP	7/0	Stable
P1.04	3ph fault @ Philipsburg 115 kV on Shawville circuitSF, normal clear loss of circuitSF and TF#2	7/0	Stable
P1.05	3ph fault @ Philipsburg 115 kV on TF#1, normal clear loss of TF#1 and Eagle Valley – Philipsburg 115 kV circuit EVP	7	Stable
P1.06	3ph fault @ Philipsburg 115 kV on TF#2, normal clear loss of TF#2 and Philipsburg – Shawville circuitSF	7	Stable
P1.07	3ph fault @ Westfall 115 kV on Eagle Valley circuitEVW, normal clear loss of Eagle Valley – Tyrone North – Westfall 115 kV circuitEVW	7/60	Stable
P1.08	3ph fault @ Westfall 115 kV on Thirty First Street circuitTFW, normal clear loss of Westfall – Thirty First Street circuitTFW	7/24	Stable
P1.09	3ph fault @ Westfall 115 kV on TF#3, normal clear loss of TF#3, TF#4	7	Stable
P1.10	3ph fault @ Westfall 115 kV on TF#5, normal clear loss of TF#5	7	Stable
P1.11	3ph fault @ Shawville 115 kV on Philipsburg circuitSF, normal clear loss of Philipsburg – Shawville circuitSF and Philipsburg TF#2	7/0	Stable
P1.12	3ph fault @ Shawville 115 kV on Garman circuitSM, normal clear loss of Shawville – Bionol Clearfield – Madera – Westover South – Garman	7/0	Stable
P1.13	3ph fault @ Shawville 115 kV on Dubois circuitSD, normal clear loss of Shawville – Rockton Mountain - Dubois, Dubois TF#3	7/0	Stable
P1.14	3ph fault @ Shawville 115 kV on TF#1A, normal clear loss of TF#1A and Unit 1	7	Stable
P1.15	3ph fault @ Shawville 115 kV on TF#2A, normal clear loss of TF#2A and Unit 2	7	Stable

Fault ID	Fault description	Clearing Time & Reclose (Cycles)	AD2-133 No Mitigation
P1.16	3ph fault @ Shawville 230 kV on Moshannon circuit MAN-SHW, normal clear loss of Shawville – Moshannon	5/0	Stable
P1.17	3ph fault @ Shawville 230 kV on Elko circuit ELK-SHW, normal clear loss of Shawville – Elko	5/0	Stable
P1.18	3ph fault @ Shawville 230 kV on Shingletown circuit SNT-SHW, normal clear loss of Shawville – Shingletown	5/0	Stable
P1.19	3ph fault @ Shawville 230 kV on TF#3, normal clear loss of TF#3 and Unit 3	5	Stable
P1.20	3ph fault @ Shawville 230 kV on TF#3, normal clear loss of TF#4 and Unit 4	5	Stable

Table 7: Single-phase Faults With Stuck Breaker (Single-Phase Delayed Fault Clear) – Category P4

Fault ID	Fault description	Clearing Time Normal and Delayed (Cycles)	AD2-133 No Mitigation
P4_1B1.01	SLG @ Eagle Valley 115 kV, normal clear loss of Eagle Valley – Philipsburg circuitEVP, SB @ Eagle Valley 115 kV, delayed clear loss of AD2-133, Q36 and Eagle Valley – Tyrone North – Westfall circuitEVW	7/20	Stable
P4_1B1.02	SLG @ Eagle Valley 115 kV, normal clear loss of Eagle Valley – Tyrone North – Westfall circuit EVW, SB @ Eagle Valley 115 kV, delayed clear loss of AD2-133, Q36, and Eagle Valley – Philipsburg circuitEVP	7/20	Stable
P4_1B1.03	SLG @ Philipsburg 115 kV, normal clear loss of Eagle Valley – Philipsburg circuitEVP, SB @ Philipsburg 115 kV, delayed clear loss of Philipsburg TF#1	7/20	Stable
P4_1B1.04	SLG @ Philipsburg 115 kV, normal clear loss of Philipsburg - Shawville circuitSF, SB @ Philipsburg 115 kV, delayed clear loss of Eagle Valley – Philipsburg circuitEVP	7/20	Stable
P4_1B1.05	SLG @ Philipsburg 115 kV, normal clear loss of TF#1, SB @ Philipsburg 115 kV, delayed clear loss of Eagle Valley – Philipsburg circuitEVP	7/20	Stable
P4_1B1.06	SLG @ Westfall 115 kV, normal clear loss of Eagle Valley – Tyrone North – Westfall circuitEVW, SB @ Westfall 115 kV, delayed clear loss of Westfall TF#5	7/20	Stable
P4_1B1.07	SLG @ Westfall 115 kV, normal clear loss of Eagle Valley – Tyrone North – Westfall circuitEVW, SB @ Westfall 115 kV, delayed clear loss of Westfall TF#3 and TF#4	7/20	Stable
P4_1B1.08	SLG @ Westfall 115 kV, normal clear loss of Thirty First Street – Westfall circuitTFW, SB @ Westfall 115 kV, delayed clear loss of Westfall TF#3 and TF#4	7/20	Stable
P4_1B1.09	SLG @ Westfall 115 kV, normal clear loss of Thirty First Street – Westfall circuitTFW, SB @ Westfall 115 kV, delayed clear loss of Westfall TF#5	7/20	Stable
P4_1B1.10	SLG @ Westfall 115 kV, normal clear loss of TF#5, SB @ Westfall 115 kV, delayed clear loss of Eagle Valley – Tyrone North – Westfall circuitEVW	7/20	Stable
P4_1B1.11	SLG @ Shawville 115 kV, normal clear loss of Shawville – Bionol Clearfield – Madera – Westover South – Garman, SB @ Shawville 115 kV, delayed clear loss of Shawville 115 kV Bus #1	7/20	Stable
P4_1B1.12	SLG @ Shawville 115 kV, normal clear loss of TF#2, SB @ Shawville 115 kV, delayed clear loss of Shawville 115 kV Bus #2	7/20	Stable

Fault ID	Fault description	Clearing Time Normal and Delayed (Cycles)	AD2-133 No Mitigation
P4_1B1.13	SLG @ Shawville 230 kV, normal clear loss of Shawville – Moshannon circuit MAN-SHW, SB @ Shawville 230 kV, delayed clear loss of Shawville 230 kV Bus #1	5/15	Stable
P4_1B1.14	SLG @ Shawville 230 kV, normal clear loss of Shawville – Elko circuit ELK-SHW, SB @ Shawville 230 kV, delayed clear loss of Shawville 230 kV Bus #2	5/15	Stable
P4_1B1.15	SLG @ Shawville 230 kV, normal clear loss of Shawville – Shingletown circuit SNT-SHW, SB @ Shawville 230 kV, delayed clear loss of Shawville 230 kV Bus #2	5/15	Stable

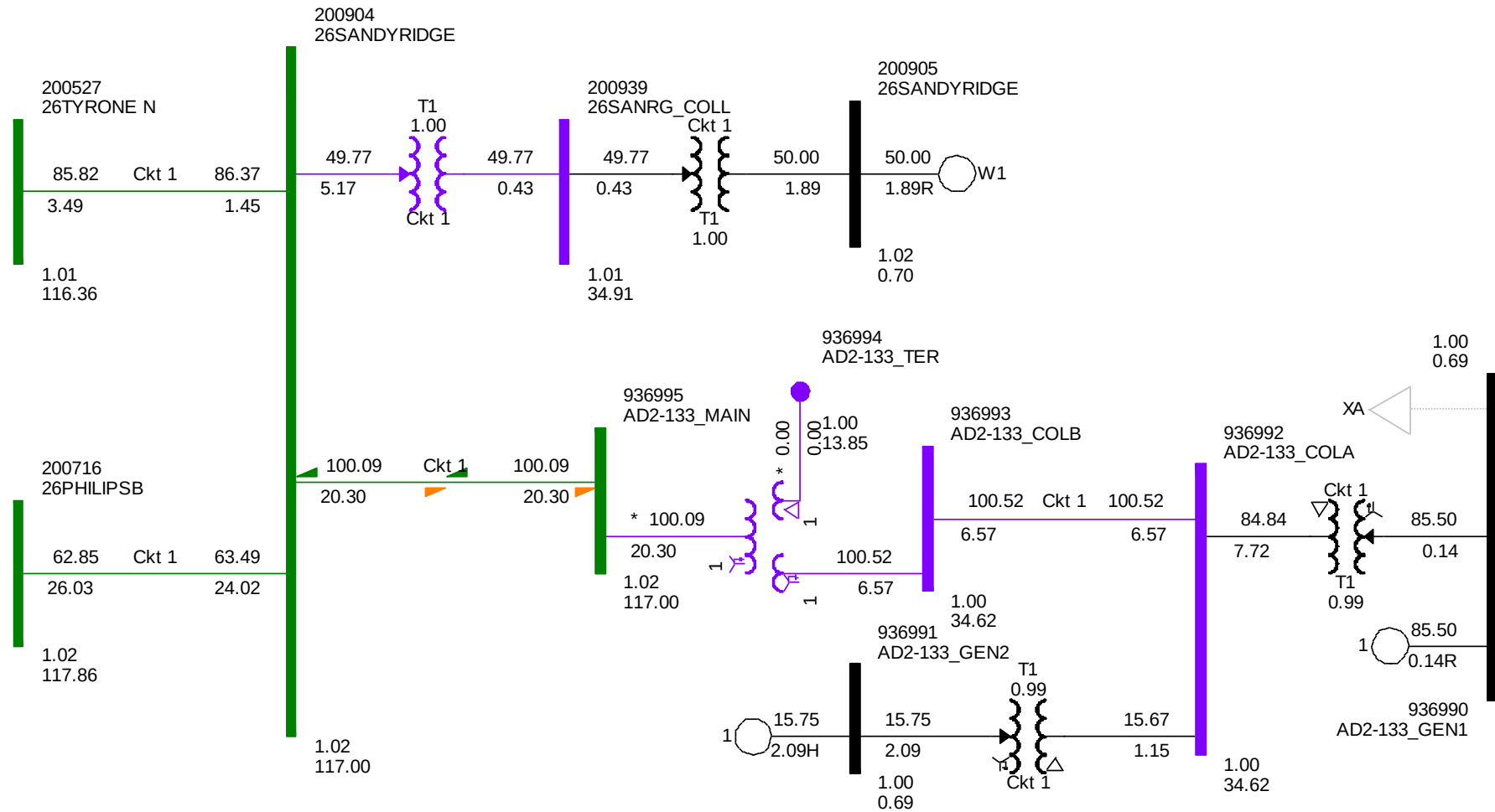
Table 8: Single-phase Faults with Delayed (Zone 2) Clearing due to Primary Communication/Relay Failure – Category P5

Fault ID	Fault description	Clearing Time (Cycles)	AD2-133 No Mitigation
P5.01	SLG fault @ 80% Eagle Valley – Philipsburg 115 kV circuit EVP, delayed clearing	7/80	Stable
P5.02	SLG fault @ 80% Eagle Valley – Westfall 115 kV circuit EVW, delayed clearing	7/80	Stable
P5.03	SLG fault @ 80% Philipsburg – Shawville 115 kV circuit SF, delayed clearing	7/80	Stable
P5.04	SLG fault @ 80% Westfall – Thirty First Street 115 kV circuit TFW, delayed clearing	7/80	Stable
P5.05	SLG fault @ 80% Shawville – Garman 115 kV circuit SM, delayed clearing	7/80	Stable
P5.06	SLG fault @ 80% Shawville – Dubois 115 kV circuit SD, delayed clearing	7/80	Stable
P5.07	SLG fault @ 80% Shawville – Elko 230 kV circuit ELK-SHW, delayed clearing	5/60	Stable
P5.08	SLG fault @ 80% Shawville – Moshannon 230 kV circuit MAN-SHW, delayed clearing	5/60	Stable
P5.09	SLG fault @ 80% Shawville – Shingletown 230 kV circuit SNT-SHW, delayed clearing	5/60	Stable

Attachment 1. Impact Study Data

Attachment 2. PENELEC One Line Diagram

Attachment 3. PSS/E Model One Line Diagram



Attachment 4. AD2-133 PSS/E Dynamic Model

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/*****
/*****
/*** Project:    AD2-133
/*** POI:       Eagle Valley 115 kV
/*** Fuel:      Wind
/*** Model:     Gamesa G114 and Siemens/Gamesa SG145
/*** Version:
/*** Size:      2.625 MW x 6 + 4.5 x 19 (100.00 MW)
/*** PSSE Version: 33
/*****

/*****
/
/ MODULE: Gamesa Type 3 Wind Turbine Generator Dynamic Data Template for PSS/E
33.11
/
/
/*****

936990 'USRMDL' 1 'REGCAU1' 101 1 1 14 3 4 1
          .02 0.7 0.9 0.5 1.2 1.2 .8 .15 -1.3 0.02 1 99 -99 .7/
936990 'USRMDL' 1 'REECAU1' 102 0 6 45 6 9
          936995 0 1 1 0 0
          0.9 1.1 .02 -.005 .005 1.1 1.0 -1.0 1.0 0.0 0 0 .02 99 -99 1.10 0.9 0.3 1.0 0.3 2.0
          1.0 0.02 0.45 -0.45 1.1 0.0 1.2 0.04 0.0 1.0 0.400 1.0 1.000 0 1.1 0.0 0.0 0.0 0.8 1.0
0.9 1.0 1.1 1.2/
936990 'USRMDL' 1 'WTDTAU1' 103 0 0 5 4 3
          10.969 0.0 0.8375 1.15 1.5/
936990 'USRMDL' 1 'WTPTAU1' 104 0 0 10 3 1
          0.5 275.0 0.5 100.0 0.0 0.3 90 -5 17 -17/
936990 'USRMDL' 1 'WTARAU1' 105 0 0 2 0 1
          .007 0.0/
936990 'USRMDL' 1 'WTTQAU1' 505 0 1 15 3 3 0

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2.0 0.01 .01 60 1.3 0.0 0.1253 0.7 0.383 0.955 0.5755 1.092 0.7844 1.12 95.0/

93699000 'FRQTPAT' 200904 936990 '1' 56.4 63.6 0.2 0.15 /
93699001 'VTGTPAT' 200904 936990 '1' 0.20 5.00 1.1 0.15 /
93699002 'VTGTPAT' 200904 936990 '1' 0.50 5.00 2.1 0.15 /
93699003 'VTGTPAT' 200904 936990 '1' 0.80 5.00 5.0 0.15 /
93699004 'VTGTPAT' 200904 936990 '1' 0.85 5.00 180 0.15 /
93699005 'VTGTPAT' 200904 936990 '1' 0.90 5.00 180 0.15 /
93699006 'VTGTPAT' 200904 936990 '1' 0.00 1.3 0.60 0.15 /
93699007 'VTGTPAT' 200904 936990 '1' 0.00 1.25 0.10 0.15 /
93699008 'VTGTPAT' 200904 936990 '1' 0.00 1.18 1.0 0.15 /
93699009 'VTGTPAT' 200904 936990 '1' 0.00 1.12 1.0 0.15 /

/*****

// SIEMENS GAMESA G114 2.625MW 60Hz Model Release: Rev 52 Variant 12
936991 'USRMDL' 1 'GMD0352' 1 1 7 60 0 500
12 0 6 1 1 1 1
14.000000 0.5000000
0.0000000 0.5000000 1.0000000 1.0000000 1.1000000 1.1000000
0.9500000 0.9500000 0.0000000 0.0000000 -0.339424 -0.339424
0.0000000 0.4500000 0.4500000 0.6500000 0.7500000 0.9000000 0.9000000
0.1550000 0.1550000 0.3000000 2.0000000 3.0000000 3.0000000 20.000000
1.2000000 1.1750000 1.1500000 1.1000000 1.1000000
0.2000000 0.5000000 1.0000000 1.0000000 20.000000
0.9633333 0.9633333 0.9633333 0.9633333 0.9633333
3.0000000 3.0000000 3.0000000 3.0000000
1.0300000 1.0300000 1.0300000 1.0300000 1.0300000
0.1000000 0.1000000 0.1000000 0.1000000
0.5000000 0.5000000
1.2100000 1.2100000 /

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/******  
/*** Project:    AD2-133 ends  
/******
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Attachment 5. AD2-133 PSS/E Case Dispatch

Bus Number	Bus Name	Id	In Service	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)	Mbase (MVA)
200503	26C.SLOPE 115.00	1	1	88	88	0	-3.2159	38.208	-17.375	102
200503	26C.SLOPE 115.00	31	1	50	50	0	-1.8272	33	-29.333	60
200665	26SHAWVL 3 22.000	3	1	175	175	120	18.3136	54	-16	201
200666	26SHAWVL 4 22.000	4	1	175	175	120	13.7338	40	-14.2	201
200715	26SHAWVL 1 18.000	1	1	122	122	62	57.1886	59	0	141
200722	26SHAWVL 2 18.000	2	1	125	125	62	57.1886	67	-45	132
200905	26SANDYRIDGE0.6900	W1	1	50	50	0	1.8912	10.1529	-10.1529	100
200913	26SHAW-D 4.1600	D	1	6	6	0	0	4	0	100
200915	26CHSTN_FL 34.500	W1	1	38	38	0	0	0	0	100
936990	AD2-133_GEN10.6900	1	1	85.5	85.5	85.5	0.1448	41.401	-41.401	95
936991	AD2-133_GEN20.6900	1	1	15.75	15.75	0	2.093	5.172	-5.172	15.75